BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 310 · Recursion and algorithm tracing

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is an activation record?

Q2. When does code after recursive call execute?

Q3. What is auxiliary stack space for depth n?

Q4. Search for 12 in [2,4,6,8,10,12,14]. Midpoint indexes?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

recursion — Subroutine calls itself.

base case — Stops additional calls.

call stack — Active subroutine frames.

unwinding — Returning from nested calls.

New learning

Understand — Trace recursive calls downward before calculating values on return.

Apply — Name the base case and variable that progresses toward it.

Evaluate — Recursion depth consumes call-stack memory even when total time is linear.

Worked example

Predict output of printing before a recursive call for n=3.

1. Explain the result — Outputs 321 on descent. Nothing prints for n0.

We Do · Practise with guidance

Predict output when printing after recursive call for n=3.

```
static void Count(int n){
 if(n==0)return;
 Count(n-1);
 Console.Write(n);
}
```

Explain the result

Outputs 123 on return as calls unwind.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace SumOdds(4), listing contributions and returned result. [6]

Question 2

For F(n): if n=0 return, else print(n) then F(n-1), state F(4) output and why. [6]

Question 3

Write an iterative C# method to sum the positive odd integers up to n (n ≥ 0), matching SumOdds(n). Compare its time and auxiliary space with the recursive method. [6]

Exit Ticket · Answer before checking

Printing after the recursive call appears on stack unwinding.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Saved data for an active function call.

Q2. On return from the deeper call.

Q3. O(n) call stack.

Q4. 3→5; index5 found.

You Do pages · Independent questions

Q1. Calls 4,3,2,1,0 [2], contributions 0,3,0,1 [2], base0 [1], result4 [1].

Q2. Print 4 [1],3 [1],2 [1],1 [1] on descent [1]; base0 prints nothing [1].

Q3. Initialise sum=0 [1]; iterate odd values using for(int i=1;i<=n;i+=2) [2]; add each i to sum and return it [1]; both approaches take O(n) time [1]; iterative O(1) auxiliary memory versus O(n) recursive call stack [1].

Exit Ticket · Model answer

True. The deeper call returns before printing.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: